

Capstone | CAP182

Part A – Related Work and Data

Machine-Learning Capstone (CAP182) · SS2 · STADIO Equities account-activation project.

Three peer-reviewed publications relevant to predicting deposit subscription / customer activation from behavioural and demographic data, followed by a description of the public dataset used to prove the approach.

Table 1 – Literature review: Moro, Cortez & Rita (2014)

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. & Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp.22-31. doi:10.1016/j.dss.2014.03.001.
Link	https://doi.org/10.1016/j.dss.2014.03.001
Modelling	Four classifiers compared under a rolling-window evaluation: Logistic Regression, Decision Tree, Support Vector Machine and Neural Network. The Neural Network gave the best discrimination (AUC ~0.80, cumulative-lift based selection). Feature relevance assessed with a sensitivity-analysis procedure.
Preprocessing	Semi-automatic feature selection reducing 150+ candidate attributes to 22; removal of the call-duration attribute for realistic (pre-call) modelling; categorical encoding; addition of socio-economic context variables (employment variation, consumer confidence, Euribor).
Datasets used	The UCI Bank Marketing dataset (Portuguese retail bank, direct telemarketing campaigns, 2008-2010). This is the same public dataset adopted in the present project.
Key takeaways	Establishes the reference formulation for predicting deposit subscription from demographic, campaign and macro-economic features. Confirms that call-duration is leakage and must be dropped for deployable models, and that macro-economic context materially improves performance - both decisions are replicated in this project.

Table 2 – Literature review: Dias & Antonio (2025)

Title	Predicting customer churn using machine learning: a case study in the software industry
Harvard reference	Dias, J.R. & Antonio, N. (2025) 'Predicting customer churn using machine learning: a case study in the software industry', <i>Journal of Marketing Analytics</i> , 13(1), pp.111-127. doi:10.1057/s41270-023-00269-9.
Link	https://doi.org/10.1057/s41270-023-00269-9
Modelling	Six supervised classifiers benchmarked: Random Forest, AdaBoost, Gradient Boosting, Multilayer Perceptron, XGBoost and Logistic Regression. XGBoost was the strongest model (Recall ~0.85, ROC-AUC ~0.86).
Preprocessing	Handling of class imbalance through resampling; feature scaling; categorical encoding; hyper-parameter tuning by cross-validated search; evaluation on a held-out set using Recall, F1 and ROC-AUC with business cost in mind.

Datasets used	A proprietary customer-behaviour dataset from a software (SaaS) provider - not public - but the customer-retention / activation framing is directly analogous to the SS1 account-activation problem.
Key takeaways	Provides recent, peer-reviewed evidence that gradient-boosted trees (XGBoost) outperform logistic regression and other ensembles on imbalanced customer-behaviour targets, and that Recall / ROC-AUC are the appropriate business metrics - supporting the choice of XGBoost as Model 2 here.

Table 3 – Literature review: Arifah et al. (2023)

Title	Application of SMOTE to handle imbalance class in deposit classification using the Extreme Gradient Boosting algorithm
Harvard reference	Arifah, D., Saragih, T.H., Kartini, D., Muliadi, M. & Mazdadi, M.I. (2023) 'Application of SMOTE to handle imbalance class in deposit classification using the Extreme Gradient Boosting algorithm', <i>Jurnal Ilmiah Teknik Elektro Komputer dan Informatika (JITEKI)</i> , 9(2), pp.396-410. doi:10.26555/jiteki.v9i2.26155.
Link	https://doi.org/10.26555/jiteki.v9i2.26155
Modelling	Extreme Gradient Boosting (XGBoost) as the classifier, combined with SMOTE over-sampling to correct class imbalance. Reported accuracy ~0.910 and ROC ~0.938 after SMOTE.
Preprocessing	SMOTE synthetic over-sampling of the minority (subscribed) class; categorical encoding; train/test partitioning; comparison of performance with and without SMOTE to isolate the effect of balancing.
Datasets used	The UCI Bank Marketing dataset (~11.7% positive rate) - the same public dataset used in this project, making its results directly comparable.
Key takeaways	Demonstrates, on the identical public dataset, that XGBoost with explicit imbalance handling is a strong performer for deposit classification. Motivates the imbalance strategy (scale_pos_weight / class_weight) used here and the recommendation to trial SMOTE as a future improvement in Part D.

Table 4 – Description of the public dataset used in this project

Name	UCI Bank Marketing - bank-additional-full.csv (the full, socio-economically enriched version)
Harvard reference	Moro, S., Cortez, P. & Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp.22-31. Dataset hosted at the UCI Machine Learning Repository. doi:10.24432/C5K306.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	41,188 rows (one row per client contact)
Number of feature columns	20 input attributes (after removing the leakage attribute duration, 19 predictors are used; the project then engineers 5 additional features, giving 77 columns after one-hot encoding)
Number of classes	2 (binary classification)
Target and its range	y - has the client subscribed a term deposit? Values: 'yes' / 'no', mapped to 1 / 0. Positive (subscribed) rate = 11.27%, i.e. an imbalanced target.
Origin	Direct (telephone) marketing campaigns of a Portuguese retail bank, May 2008 - November 2010, enriched by the authors with five Banco de Portugal socio-economic indicators (employment variation rate, consumer price index, consumer confidence index, Euribor 3-month rate, number of employees).

Note: two of the three reviewed publications (Moro et al., 2014 and Arifah et al., 2023) use this exact public dataset, satisfying the requirement that at least one reviewed paper relies on a dataset also available to this project and enabling direct comparison of results in Part C and Part D.